Machine Learning and Deep Learning: Two Real-World Applications

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Machine learning (ML) enables computers to learn patterns from data and make predictions. Deep learning (DL) is a subset of ML that uses multilayer neural networks to learn complex representations. This report examines two examples from the lesson: house price prediction using multiple linear regression and image recognition using convolutional neural networks (CNNs). These applications illustrate how data characteristics, feature extraction, and practical requirements influence the selection of an appropriate approach.

Traditional Machine Learning: House Price Prediction

A research application of traditional ML is Xu's (2023) study of housing prices in Ames, Iowa. Using real housing data, Xu constructed multiple linear regression models with predictors including overall material and finish quality, above-ground living area, garage capacity, garage area, and basement area. The study also examined how removing closely related predictors affected model fit, demonstrating the importance of selecting appropriate variables.

Multiple linear regression suits this problem because the inputs are structured property characteristics. Researchers can select relevant features before training and estimate their relationships with sale prices. Unlike image recognition, the task does not require discovering visual representations from raw pixels. Regression coefficients also allow users to examine how each included feature relates to predicted prices while holding other variables constant.

However, interpretability does not guarantee accuracy. Regression coefficients represent associations rather than proving causation. Correlated predictors, omitted characteristics, and

changing market conditions can affect predictions. Therefore, a model should be evaluated on unseen sales rather than judged only by how closely it fits its training data.

For this particular application, deep learning may be less suitable because additional complexity is not necessarily justified by the selected inputs. A neural network would introduce more modeling choices and potentially greater computational costs. Without sufficient data or complex patterns to learn, it could overfit rather than improve prediction. This comparison is an analytical assessment, not a finding established by Xu's study. Deep learning could become more appropriate if the task incorporated extensive property photographs or other unstructured data.

Deep Learning: Retinal Image Recognition

A real-world application of image recognition is detecting diabetic retinopathy from retinal photographs. Gulshan et al. (2016) trained a deep CNN using 128,175 images evaluated by medical professionals. The model demonstrated high sensitivity and specificity for detecting referable diabetic retinopathy on separate evaluation datasets.

Subsequent research examined deep-learning screening in practice. Ruamviboonsuk et al. (2022) evaluated a system at nine primary care sites in Thailand. It provided image interpretations and referral recommendations in real time, while retinal specialists reviewed images as a safety measure. This study connected deep-learning image analysis with an actual healthcare workflow.

A CNN suits this problem because retinal photographs contain complex spatial information. Convolutional layers learn visual features during training, and successive layers

combine simpler patterns into representations useful for classification. This reduces the need to manually define every potentially relevant image characteristic.

Traditional ML could also classify retinal images, but it would generally require a separate feature-engineering process. Developers might extract measurements of lesions or textures before training a classifier. Such predefined features could miss useful information or respond inconsistently to changes in imaging conditions. Consequently, traditional ML may be less suitable when the task requires learning numerous subtle visual patterns.

Nevertheless, deep learning has limitations. Training requires labeled images and computational resources, while image quality and population differences can affect performance. Ruamviboonsuk et al. (2022) also emphasized the importance of implementation conditions and workflows. Strong model performance therefore does not eliminate the need for validation and appropriate clinical oversight.

Conclusion

These applications demonstrate that model selection should reflect the problem rather than a preference for complexity. Multiple linear regression provides an interpretable approach to structured housing data, while CNNs learn representations suited to complex retinal images. Data requirements, computational costs, interpretability, and performance on unseen examples should guide the final choice.

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